

Comparing small area estimation methods for poverty indicators in the municipalities of Minas Gerais State

Solange Correa (s.correa-oneil@soton.ac.uk)¹, Debora Souza², Nicia Brandolin², Viviane Quintaes² and Djalma Pessoa²

Keywords: Small Area Estimation, Poverty Indicator, Inequality.

1. INTRODUCTION

Official statistical agencies have been facing growing demand for detailed and accurate information and, on the other hand, suffer from constant financial constraint in the production of sample surveys. In this sense, researchers have been studying methods for small domain estimation in order to provide estimates for small geographic areas or domains with controlled precision without increasing costs.

Methods to estimate poverty in small areas combine information collected from multipurpose household sample surveys with information from censuses with broad geographic coverage. The poverty map published in 2008 by the Brazilian Institute of Geographic and Statistics [1] adopted the method proposed by Elbers *et al.* [2], the so called ELL method, and aimed to give a detailed description of the spatial distribution of poverty in the country. More recently, Molina and Rao [3] have developed a methodology (MR approach) that uses as much sample information for estimation of poverty indicators in small areas as possible. According to the authors, this proposal results in lower mean square errors compared to the ELL approach.

The main objective of this work is to compare in terms of bias and mean square error (MSE) the methods proposed by Elbers *et al.* [2] and Molina and Rao [3]. A simulation study is conducted using data from the 2008-2009 Brazilian Budget and Expenditure Survey (POF) [4].

2. METHODS

Consider a finite population of size N partitioned into D areas of sizes $N_1, \dots, N_D$. Suppose that E_{dj} is an adequate measure of welfare, e.g. income or expense, for individual j in small area d . Let z be a given poverty line, i.e. the threshold for E_{dj} below which a person is considered poor. The *FGT* family of poverty indicators for each small area d [5] is given by

$$FGT_{cd} = \frac{1}{N_d} \sum_{j=1}^{N_d} \left(\frac{z - E_{dj}}{z} \right)^c I(E_{dj} < z), \quad c = 0, 1, 2; \quad d = 1, \dots, D, \quad (1)$$

¹ University of Southampton, UK

² Brazilian Institute of Geography and Statistics (IBGE)

where $I(E_{dj} < z) = 1$ if $E_{dj} < z$ (person is considered poor) and $I(E_{dj} < z) = 0$, otherwise.

For $c = 0$, expression (1) becomes the poverty rate or poverty incidence in area d . When $c = 1$, the FGT measure is called the poverty gap, i.e. the average relative distance to non-poverty for individuals in area d . When $c = 2$, expression (1) is called poverty severity and emphasises the extreme poverty.

The ELL method consists in estimating social welfare for the same geographic level of the Demographic Census. This information is obtained directly from external sources or derived from household sample surveys. As a result, it is possible to use the geographically detailed information from the Census to estimate poverty indicators and corresponding variances for small domains. The idea of the MR approach is to combine household survey data to Census data and apply random intercept two-level models to predict the welfare variable.

To compare the ELL and MR approaches, samples are selected from the Demographic Census data using the same complex sampling design used in the 2008-2009 Brazilian Budget and Expenditure Survey (POF). For each sample, poverty incidence and poverty gap are estimated by both methods and compared with those calculated in the population. This study may provide subsidies to choose an appropriate methodology to produce a poverty map for the country.

3. RESULTS AND CONCLUSIONS

In the simulation study, 400 random samples are selected from the 2010 Demographic Census data for the Minas Gerais State in Brazil using the same sampling design adopted by POF. Both methodologies, ELL and MR are applied to each sample and estimates of poverty measures (poverty incidence and poverty gap) obtained. These estimates were then compared in terms of bias and MSE with the quantities values calculated directly from the Census data. The small area of interest in this study is the municipality and the variable of welfare considered is the *per capita* household income.

Table 1 and Table 2 show descriptive statistics for the relative MSE and relative bias of the poverty rate and poverty gap, respectively, for municipalities of Minas Gerais State. Results show that both the relative bias and the relative MSE a higher for municipalities with lower values of poverty incidence. In addition, both methods overestimate the poverty rates, with the MR approach producing higher relative bias and relative MSE. Similar pattern was observed for the poverty gap.

Table 1 Descriptive statistics for the relative MSE and relative bias of the poverty rate for municipalities of Minas Gerais State.

Statistic	Estimate (%)			Relative MSE (%)		Relative Bias (%)	
	Population	ELL	MR	ELL	MR	ELL	MR
Minimum	7.32	11.29	9.81	0.04	0.05	-24.04	-21.38
1st Quantile	18.02	22.64	25.68	0.46	1.06	3.13	8.15
Median	26.99	29.92	32.16	1.86	4.27	12.52	19.86
Mean	29.07	32.16	34.3	7.71	15.39	17.42	27.49
3rd Quantile	39.15	40.5	41.92	8.06	16.98	27.62	40.27
Maximum	65.25	65.46	66.39	186.24	318.55	133.93	176.49

Table 2 Descriptive statistics for the relative MSE and relative bias of the poverty gap for municipalities of Minas Gerais State.

Statistic	Estimate (%)			Relative MSE (%)		Relative Bias (%)	
	Population	ELL	MR	ELL	MR	ELL	MR
Minimum	1.93	3.46	3.22	0.12	0.07	-45.15	-36.11
1st Quantile	6.21	7.39	9.19	0.68	1.12	-7.99	3.82
Median	10.35	10.54	12.38	2.03	4.62	4.22	19.73
Mean	12.41	12.32	14.05	7.88	23.94	9.52	30
3rd Quantile	17.25	16.34	17.92	5.83	23.89	21.11	47.77
Maximum	40.26	35.78	36.81	350.13	713.31	182.83	261.6

This is work in progress and extensions include comparison of the variance estimates obtained under both approaches.

REFERENCES

- [1] IBGE, Mapa de pobreza e desigualdade: municípios brasileiros 2003, Rio de Janeiro: IBGE (2008).
- [2] C. Elbers, J. Lanjouw and P. Lanjouw, Micro-level estimation of poverty and inequality, *Econometrica* 71 (2003), 355-364.
- [3] I. Molina and J. Rao, Small area estimation of poverty indicators, *Canadian Journal of Statistics* 38 (2010), 369-385.
- [4] | IBGE, Pesquisa de Orcamentos Familiares 2008-2009 Despesas, Rendimentos e Condições de Vida, Rio de Janeiro: IBGE (2010).
- [5] J. Foster, J. Greer and E. Thorbecke, A class of decomposable poverty measures, *Econometrica* 52 (1984), 761-766.